ConTact Sensors: A Tactile Sensor Readily Integrable into Soft Robots

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***Abstract*— The compliant nature of soft robots make them attractive for robotics, especially in applications involving human interaction or unstructured environments. However, soft robots are often complex to control and present difficulties in determining their state. External sensing, such as optical tracking, can be used for state estimation, but is often cumbersome. Internal sensors built into the robot can also be used for feedback but conventional sensor designs are primarily intended for rigid structures. Soft robots require soft sensors that are more akin to their stiffness. Ultimately, soft robots should use inherent sensing capabilities found in their actuation fluids and deformable structures. In this paper, we propose the concept of using multiple modalities of sensing, fluid pressure and electrical resistance, of the fluid medium. Each modality responds differently to the same stimulus, allowing richer sensing than two distinct sensors would provide. We embody this concept into ConTact Sensor, a soft tactile sensor that can detect the force and contact area of the object compressing it. ConTact Sensor shares similar fabrication materials as most soft robotic designs, thus it can be combined with soft actuators or even embodied into a soft actuator itself. We detail the design, fabrication and characterization of ConTact Sensor and provide an example of integrating the sensor in a liquid-filled PneuNets actuator.**

I. Introduction

Control in soft robotics has primarily relied on open-loop control. This can be forgiving in some use cases, such as a soft robotic gripper that takes advantage of its natural compliance. In some cases, such as robotic walkers, a simple open-loop replay is enough to control the robot for a certain performance envelope. [1] In more complex scenarios, such as dexterous manipulation or robots that interact with the environment, the dynamics are unpredictable and control is more difficult, and would greatly benefit from having feedback [2][3].

A. Sensors and State

In rigid robotics, feedback usually comes from straightforward sensors that neatly correspond to degrees of freedom, such as load cells and encoders. State estimation is then a matter of incorporating sensor estimates of each degree of freedom into a model. It is not straightforward to apply this strategy to soft robots, however, because they typically have intractably high degrees of freedom, therefore some soft robots rely on external feedback, such as optical trackers [4].

However, relying on external sensing often results in limitations in environments and work volumes. For example, a snake-like robot that relies on optical trackers cannot be used in confined or visually occluding spaces, an environment in which it would definitely excel in. One concept to solve this issue is to embed exclusive structures designed for sensing in the robot, such as resistive flex sensors [5], and hall effect sensors [6][7]. However, this form of feedback comes at the cost of a mismatch between stiffnesses of the sensor and the robot, therefore any sensors built into the robot must themselves be soft.

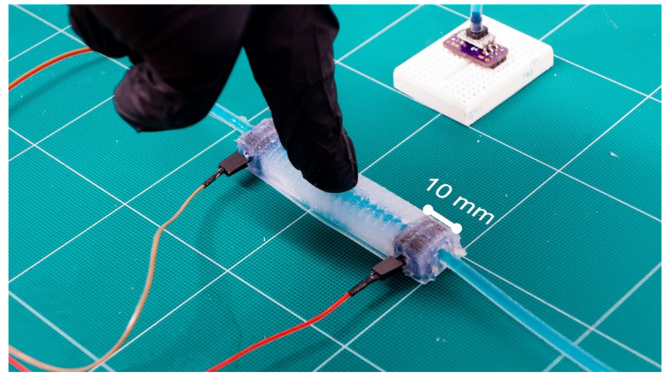

Fig. 1. ConTact Sensor being used as a tactile sensor to determine the force and contact length of an experimenter's fingertip. The name ConTact is from the main principle of sensing: a Conductive fluid, and the sensor itself being a Tactile sensor.

Another approach is to integrate softer materials to be used as sensors. Some integrated sensors are based on elastomers built into the soft robot that include channels for liquid-phase materials [8]. The elastomer layers are patterned with channels for gallium alloys [9], designed so that when the channel shape changes, a desired physical property (e.g. curvature of the robot) is sensed. Similarly, optical channels can be used to detect deformation of the waveguide, and to relate that change to the state of the robot [10]. More recently, embedded 3D printing was used to integrate multiple distinct sensors of different properties (curvature, contact, inflation, etc.) into the same pneumatically-actuated soft robotic system [11].

B. Sensing the Actuated Medium

Building separate actuation and sensation subsystems into the soft robot, however, fails to take advantage of the fact that the actuated fluids or materials of the robot are themselves potential sensors. Recently, conductive fluid (salt water) was used as the actuation fluid for a flexible fluidic actuator [12], which enabled primitive proprioception to sense the angle of the bending actuator. 3D printed soft sensors based on the electrical resistance of photopolymers [13] were

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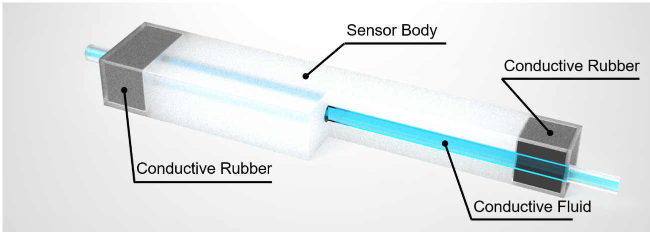

Fig. 2. ConTact Sensor's components in a cross-section view. The sensor body is depicted as cut away at the halfway point for better visualization of the conductive fluid channel.

also proposed which can enable tight integration into the fabrication of soft robots. These techniques are still emerging aspects in the soft robotics community.

C. Sensing Multiple Properties

In this paper, we describe a means for sensing more than one dynamic characteristic of the fluid medium, namely the electrical resistance and fluid pressure of a conductive hydraulic fluid. We propose a tactile sensor design that is able to discern between a range of physical depressions from light to heavy and from small to large that is able to be integrated into most existing soft robot designs. The concept is not exclusive to adding sensors to soft robotics, but can also be embodied into the whole soft robot itself, enabling proprioception and tactile sensing within a single medium.

II. Design & Fabrication

In its essence, ConTact Sensor is ionic solution housed in a deformable tube. The sensed quantities are the changes in internal pressure and the changes in the electrical resistance of the fluid, which according to the resistivity formula ($R = \frac{\rho L}{A}$), the resistance (R) given a constant resistivity (ρ), depends on the length (L) and cross-sectional area (A) of the medium. If the sensor is pressed into or pinched, the cross sectional area would decrease leading to an increase in electrical resistance. The resistance also varies with the amount of length-wise area being affected. And in both cases, the overall fluid pressure in the sensor increases. Using this duality of sensing characteristics, we can derive richer sensing capabilities from a single structure.

ConTact Sensor is designed to be readily integrable into traditional silicone-based soft robots therefore the sensor is primarily fabricated from casting soft silicone elastomer (Platsil Gel-25, Polytek Development Corp.) in a plastic mold. An electrically conductive interface is required between the measurement equipment and the fluid to obtain the electrical resistance. However, simply inserting electrical wires to contact the fluid through the elastomer would certainly be prone to leakages [14], therefore conductive silicone fabricated from silicone elastomer and chopped carbon fibers is used as the interface. Since the conductive silicone is made from the same type of elastomer as the body, the sensor's fluids are sealed from the outside environment and the sensor is significantly less prone to leakage. Another benefit of using conductive silicone rubber is reducing the transition between low-stiffness materials e.g. the silicone rubber and fluids to high-stiffness materials e.g. the copper wires, which should improve the mechanical robustness of the sensor.

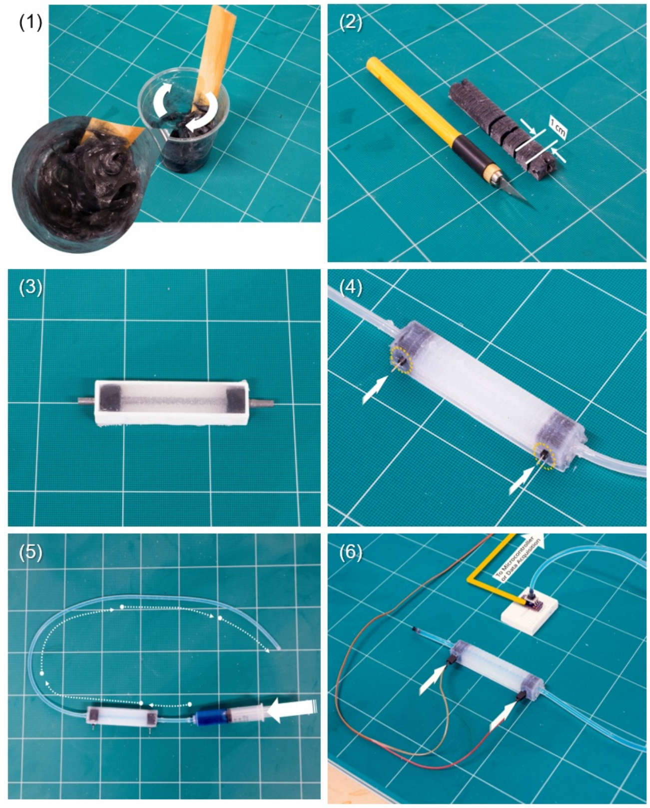

Fig. 3. Overview of the steps for fabricating ConTact Sensor, as numbered; 1. Preparing the conductive silicone rubber 2. Cutting the conductive rubber into 10 mm long sections 3. Casting the main body of the sensor with the conductive rubber in-line with the metal rod. 4. Adding copper leads to the sensor. 5. Filling the sensor with salt water dyed with food coloring. 6. Attaching ConTact Sensor to the data acquisition system.

ConTact Sensors can be produced in any shape and size as long as three main components are present: the main sensor body, the conductive rubber ends, and the ionic solution, as shown in Fig.2. For this proof of concept we fabricated a main sensor body in a brick-like shape, 16 mm wide by 10 mm tall by 85 mm in length, with a 4 mm diameter tubular section spanning the length of the whole body. The conductive rubber ends, which conduct electrical current between the internal fluid and the exterior, are 10 mm long rectangular sections with a 4 mm hole embedded in the sensor's body at each end. The ionic solution is simply saline solution with 5 parts table salt to 95 parts plain water by weight (5 %wt NaCl).

The fabrication of ConTact Sensor can be divided into three main steps: casting the conductive silicone ends, casting the sensor body and filling the sensor prior to use. A comprehensive guide to fabricating ConTact Sensors is also available at the Soft Robotics Toolkit website. A brief overview of all the steps involved is shown in Fig.3 and the following are the fabrication steps in detail and reasoning behind them.

A. Conductive Silicone Rubber

Commercially available conductive silicone elastomer are often propriety mixes of silicone rubber with graphite particles, which have long cure-times and less than ideal mechanical properties. [15] The conductive silicone rubber in ConTact Sensor is made by mixing 5 mm long carbon fiber strands (.125" Chopped Carbon, Innovative Composite Technologies) with two-part platinum-cure silicone rubber (Platsil Gel-25, Polytek Development Corp.). The resulting carbon fiber and silicone rubber mixture cures just as fast as the bare silicone rubber and the cured conductive rubber is compliant and not brittle. The method was first published by Andrew Quitmeyer on the Instructables website, initially intended for use in pleasure devices. [16]

Casting the conductive silicone rubber for one ConTact Sensor starts with soaking 0.5 g of chopped carbon fiber strands in 1 g of 70 % isopropanol, which is just enough to wet the strands and disperse them. The wet strands are then either left to partially dry or squeezed to remove the excess alcohol. The strands are then vigorously stirred into a mixture of 20 g of Part A and 20 g of Part B of the silicone rubber. The carbon fiber-silicone rubber mixture's conductivity can be verified at this step using an ohmmeter. The ratio of 0.5 g to 40 g silicone elastomer was found empirically through trial-and-error and has desirable viscosity for casting and adequate electrical conductivity —in the order of several hundred Ohms in resistance across 10 mm. Increasing the amount of carbon fiber strands can decrease the overall resistance at the cost of higher viscosity, which could lead to problems during pouring and casting. Before casting, a 4 mm diameter metal rod is inserted into the mold and finally the now paste-like mixture is spread into the mold. The metal rod serves to displace the volume that would become the 4 mm inner channel of the sensor. A minimum time of one hour is required for the room temperature vulcanization (RTV) of the rubber to complete. Once the rubber has dried, an ohmmeter is used to verify the electrical resistance of the conductive rubber. The conductive rubber is then removed from the mold and cut into 10 mm long sections.

B. Sensor Body

ConTact Sensor's main body is cast from a similar platinum-cure silicone rubber (Plat-Sil Gel 10, Polytek Development Corp.) to the conductive rubber by using the same mold. The conductive ends are positioned in a metal rod within the mold. Then, the silicone rubber is simply poured into the open-top mold and left to vulcanize. Once the rubber has dried, the metal rod is removed leaving a 4 mm diameter hollow section throughout the sensor. To complete the sensor, two 4 mm silicone tubes with an inner diameter of 2 mm (PUTC4, MiSUMi Group Inc.) are glued in place with silicone adhesive. (Sil-Poxy, Smooth-On, Inc.) The same adhesive is used to hold copper pins that have been pierced into the conductive rubber in place. Note that the silicone adhesive is only for holding the pins in place, and does not serve any purpose to seal the sensor.

C. Conductive Fluid

The conductive fluid used in the ConTact Sensors presented in this paper is 5% saline solution or 5 wt% Sodium Chloride (NaCl). The solution is prepared by weighing 5 grams of table salt (The Kroger Co.) followed by 95 grams of room temperature tap water. Food coloring (The Kroger Co.) is added to make the fluid visible through the semi-transparent silicone rubber and tubing.

Alternative ionic fluids, such as 1-ethyl-3-methyl imidazolium ethylsulfate (EMIM-ES) or salt dissolved in glycerin [17][18][14], could possibly be used, but salt water is trivial to acquire and prepare. A lower concentration akin to medicinal saline (0.9% w/v) could also be used for ConTact Sensors in contact with humans as it is considered roughly isotonic to human cells. [19]

As we are measuring the fluid's electrical resistance, one major drawback of using salt water as the conductive fluid is the solution is ionic, which means that the resistance cannot be measured with the standard method of applying a DC voltage and measuring the voltage drop. Applying a DC voltage across two leads in salt water will cause the positive ions (Na^+) to pool on the negative lead and negative ions (Cl^-) to pool on the positive lead, causing a charge buildup which gives an erroneous reading of the resistance.

In this paper, we use alternating current (AC) to measure the electrical resistance. Using our data acquisition system (USB-6251, National Instruments) we apply a 1 volt peak-to-peak sinusoidal waveform centered at 0 volts across the ConTact Sensor and a known resistance in a voltage divider configuration as shown in Fig.4. The resistance of the fluid inside the sensor can then be calculated as (1).

$$R_{ConTact} = R_{known} \times \frac{V_M}{V_M - V_S} \tag{1}$$

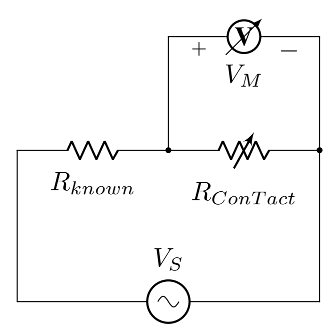

Fig. 4. The voltage divider circuit used for measuring the salt water with a sinusoidal waveform.

III. Sensor Characterization

A. Material Properties

The carbon fiber based conductive silicone rubber had a volume resistivity ranging from 750 to 2460 $\Omega \cdot m$ across 6 different batches with an average of 1340 $\Omega{\cdot}m$. However, the overall resistance of each distinctly shaped sample remained

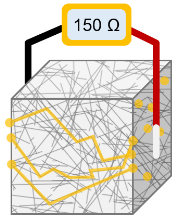

Fig. 5. An abstraction of the hypothesized conduction paths for the conductive rubber. The individual carbon fiber strands, represented as gray lines, are in contact with each other in random orientations as suspended by the silicone rubber, represented as the off-white cuboid. The contacting strands make a conduction path from one face of the cuboid to other faces, as represented by the yellow lines.

near 150 Ω. We hypothesize that the conductivity of the silicone is from the random orientation of carbon fiber strands creating a network of conductive paths propagating throughout the silicone rubber, as shown in Fig.5.

B. Experimental Methods

We performed a series of indentation experiments in order to calibrate and characterize the sensor. The sensor was placed on a custom platen on a universal testing machine (ElectroForce 3200, TA Instruments). The universal testing machine is designed for testing soft and compliant specimens using forces lower than 200 N. The platen is mounted on top of a load cell (WMC-50, Interface Inc.). The setup can be seen in Fig.6 The cylinder of the testing machine is fitted with indenters of various widths and shapes, as shown in Fig.7 ranging from 2.5 mm to 20 mm in width.

The sensor's conductive rubber ends were connected to our data acquisition system. One end of the silicone tubing was connected to a pressure sensor (ABPMANT100PG2A3, Honeywell) that interfaces via I^{2}C with a microcontroller (Teensy 3.5, PJRC) and the other end was capped shut. We performed all data logging using LabVIEW (National Instruments) software to record the fluid electrical resistance, pressure, force from the loadcell and indenter displacement.

Each experiment consists of moving the indenter down to a depth of 5.5 mm, starting directly on top of the sensor, with a fixed speed of 2 mm/s, then dwelling for three seconds and finally lifting up to its starting position.

C. Experimental Results

We performed 5 indentation trials per indenter with a total of 7 indenters, totalling 35 trials. We preprocessed the data by truncating the data stream for each indentation to right before the indenter starts moving down and right after the indenter returns to its starting position. ($\sim$12 seconds) The data was collected at a sampling frequency of 50 Hz. The sensor's electrical resistance data in voltages was fed through a third order median filter and then converted to Ohms using (1) with R_{known} set to 20 $k\Omega$. The resistance was then scaled to the starting resistance of each trial and zeroed, achieving $\Delta R/R_0$ as shown in subsequent plots. Displacement, pressure and force data was simply scaled to their pre-calibrated values and then zeroed to each trial.

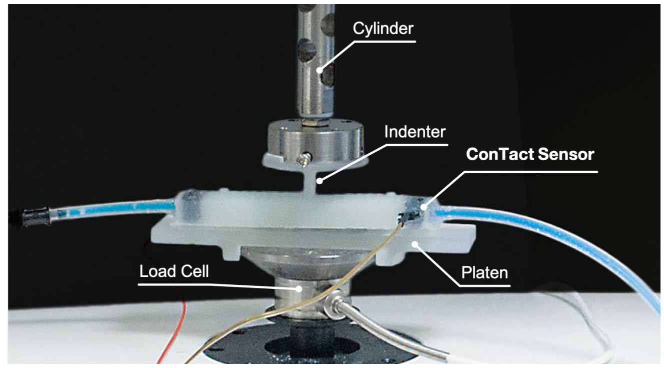

Fig. 6. The experimental setup: ConTact Sensor is placed on a 3D printed platen that is mounted on a load cell. The material testing machine's cylinder is fitted with an 3D printed indenter. The background has been artificially darkened for clarity.

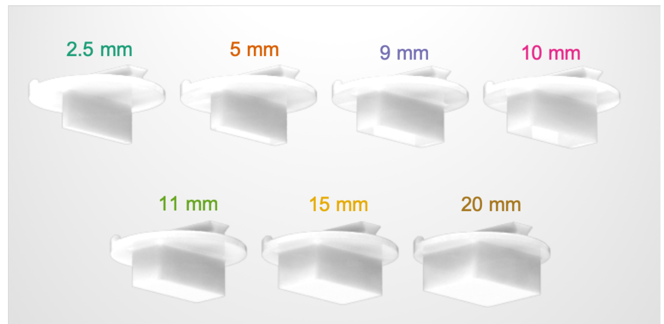

Fig. 7. Indenters with their widths printed directly above, color coded to match subsequent plots in this paper.

As shown in Fig.8 (a), when plotting the electrical resistance ($\Delta R/R_0$) against the fluid pressure (ΔP), the sensor clearly responds differently to each size indenter. As the indenter starts loading into the sensor, at first the pressure increases, after a certain point the resistance also increases, at a faster rate, and remains stable when the indenter has reached its lowest position. We hypothesize that due to the nonlinear response to cross-sectional area change (or pinching), the resistance barely shows any response during the "pressure sensitive" period, but rapidly catches up during the "transition" period which is just prior to the pressure saturating. During the "resistance sensitive" period, the sensor responds almost exclusively to the change in cross-sectional area or local collapse of the cavity. It can be observed from Fig. 8 (b) and (c) that with pressure data alone, the sensor cannot discern between a 9 mm and 10 mm indenter.

D. Estimation

Using data only from the duration of the downstroke of the indentation experiments, we fit a linear regression model to the pressure and resistance data to predict the force acting on the sensor. We used MATLAB's `fitlm` to perform linear regression using ordinary least squares weighting and allowing for interactions between input variables. The resulting model (2) predicts the force to within $\pm$12.28 N of the actual force, which is approximately 10% of the full dynamic range.

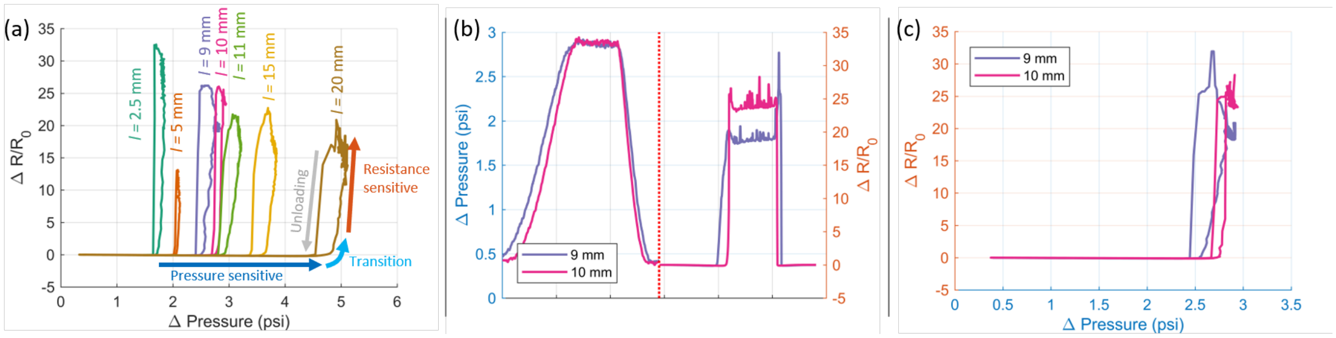

Fig. 8. ConTact Sensor's response to various indentation sizes and forces. (a) Plot showing the color-coded raw responses to indenters shown in Fig.7. The sensor shows a transition of behavior from responding in pressure increase to responding in resistance increase. (b) yy-axis plot showing almost identical pressure mode response for different sized indenters, but significantly different response in resistance mode. (c) The same data from (b) when plotted in a Pressure-Resistance plane.

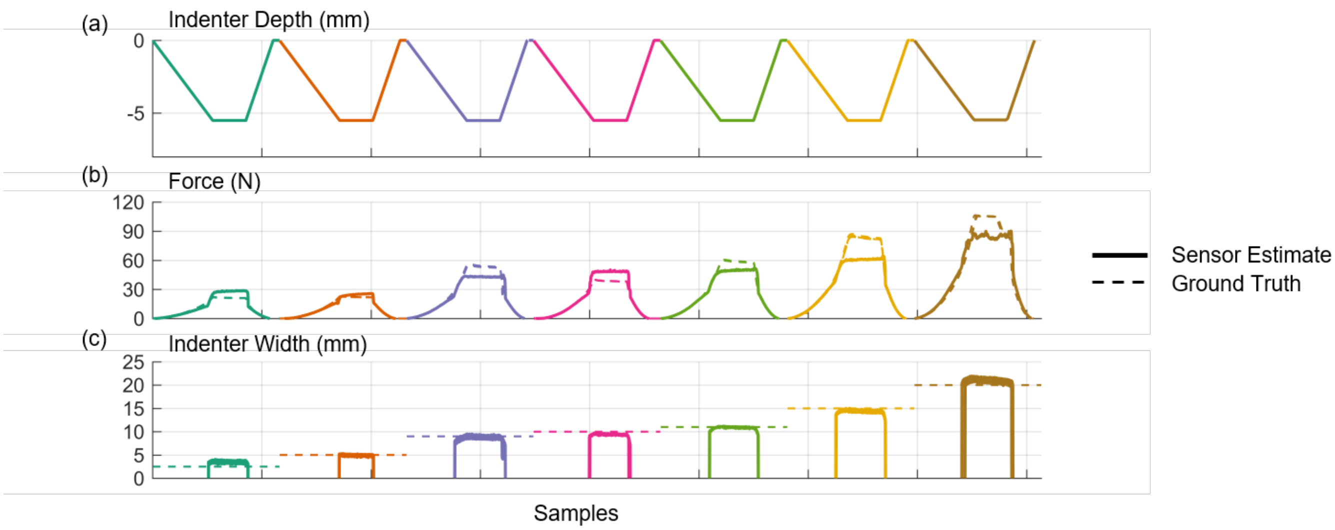

Fig. 9. Plots showing the force and contact length estimation derived from the sensor readings. Each section represents a different width indenter. (a) Plot of the indenter's depth into into the sensor (b) Plot of the force estimation and ground truth. (c) Plot of the contact length estimation and ground truth.

$$F = k_r R + k_p P + k_i RP + k_{r2} R^2 + k_{p2} P^2 \tag{2}$$

Looking at the sensor response in Fig.8 (a), for each sized indenter, the sensor's resistance rapidly increases at different pressures. Knowing this, we used the resistance and an empirically derived threshold as an activation function for the pressure. In other words, the contact length is zero until a certain resistance is reached, and then the contact length is the pressure scaled by some value. Using the model, we can estimate the contact length to within 0.429 mm of the actual length.

$$L = (k_l P + L_0) \times 1(R \geq R_{thresh})$$
$$1(R \geq R_{thresh}) = \begin{cases} 1, & R \geq R_{thresh} \\ 0, & R < R_{thresh} \end{cases} \tag{3}$$

IV. Application

As a demonstration of ConTact Sensor's ability to be integrated into soft robotic designs, we embedded one into the tip of a PneuNets Actuator, as shown in Fig.10(a). We designed the actuator with the intention of having the sensor in the same hydraulic circuit as the actuator, i.e. the actuation fluid passes through the sensor during inflation and deflation. The actuation fluid was 5%wt salt water, prepared in the same manner as mentioned in the previous sections. To achieve the complex geometry, the actuator was cast using lost-wax casting.

We performed a series of experiments by pressing the sensor-actuator hybrid against six indenters from Fig.7 (sizes 2.5 mm, 5 mm, 9 mm, 10 mm, 11 mm and 15 mm). The results show that the embedded ConTact Sensor can discern between different sized indenters, as shown in Fig.10(c). The overall response is similar to the results in the previous section, but a linearized model of the response is more difficult to derive. We expect to expand this result in the near future and apply learning-based predictions in place of the regression.

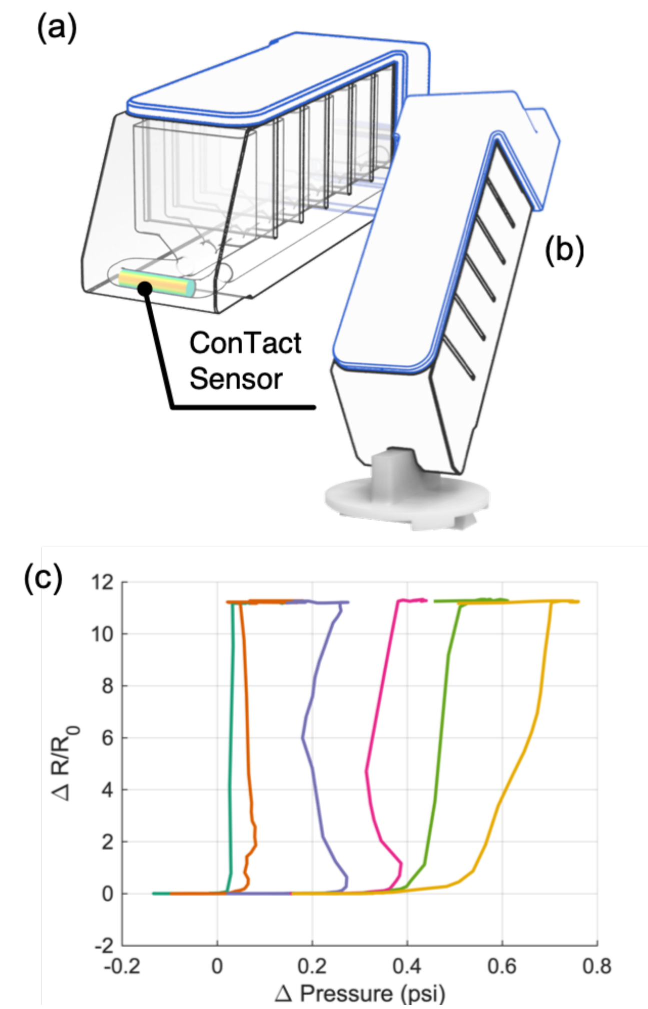

Fig. 10. ConTact Sensor's response when embedded in the tip of a modified PneuNets Actuator. The sensor-actuator hybrid can discern small from large contact lengths.

V. Conclusion & Future Work

Arguably, the ConTact Sensor is simple in concept and design. However, the sensor's simplicity enables it to be integrated into other soft robotic designs, especially those with liquid-based actuation. We presented a novel method of sensing force and contact area by using a single fluid medium and the overall concept of using more than one physical characteristic of sensing medium to achieve richer sensing.

Future work will be on expanding the modes of sensing, as of now ConTact Sensor primarily responds to normal force, but with complex geometries of the inner channel we hope to be able to measure tangential force (shear force). As discussed in the previous section, the sensor output is more challenging to interpret when it is integrated with a PneuNets actuator. We will continue to refine the design until full proprioception is achieved.

Acknowledgement

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